

State Fiscal and Regulatory Policy Effects on Economic Performance

Evidence from U.S. State-Level Panel Data, 2000-2020

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Executive Summary

Research question	How do state fiscal and regulatory policies relate to state economic performance across U.S. states?
Sample	Balanced state-year panel: 50 states over 2000-2020; N = 1,050 observations.
Outcomes	Real gross state product per capita and real private-sector gross state product per capita.
Estimator	State fixed-effects regressions using fiscal and regulatory policy variables from Cato Institute, BEA, and Census-based population data.
Main takeaway	Fiscal variables were more consistently associated with state economic performance than most regulatory variables tested; several regulatory variables were not statistically significant in the fixed-effects specifications.

This portfolio version revises the original course paper into a cleaner research sample with additional technical detail. It is intended to show the empirical workflow, modeling choices, and policy interpretation behind an undergraduate econometrics project presented outside the classroom.

1. Motivation and Research Design

The U.S. federal system creates substantial cross-state variation in taxation, public spending, regulatory institutions, and market rules. This project uses that variation to examine whether states with different fiscal and regulatory policy environments exhibit different levels of economic output per person. The analysis is motivated by a core public finance question: when state governments tax, spend, regulate, or shape local institutional incentives differently, are those differences associated with measurable differences in economic performance?

The project does not claim to estimate a fully causal structural model. Instead, it uses state fixed-effects regression to make a disciplined empirical comparison within states over time. This is a meaningful improvement over a purely cross-sectional comparison because it accounts for time-invariant differences across states - such as geography, long-standing industrial composition,

climate, inherited institutional culture, or persistent regional economic advantages - that could otherwise confound estimates.

2. Data Sources and Construction

The dataset combines state-level policy variables with state economic output and population data. The final working sample covers 50 states from 2000 through 2020, producing 1,050 state-year observations. Each observation corresponds to one state in one year.

Component	Source	Role in analysis
Policy variables	Cato Institute, Freedom in the 50 States	Fiscal, regulatory, and institutional policy variables used as regressors.
Economic output	Bureau of Economic Analysis regional economic accounts	Real gross state product and private-sector gross state product.
Population	U.S. Census state population estimates	Denominator used to create per-capita outcome variables.

Outcome variables

The analysis uses two dependent variables: real GSP per capita and real private-sector GSP per capita. Per-capita measures place states of very different population sizes on a more comparable scale. The second outcome narrows the focus to private-sector output, which is useful because fiscal and regulatory policy may affect the private economy differently than total state output.

Policy variables

The regressors include fiscal variables such as state government consumption and investment relative to income, state and local tax burdens, state debt relative to income, state and local public payrolls relative to private payrolls, and aggregate fiscal policy indexes. The regulatory variables include measures related to regulatory takings compensation and assessment rules, eminent domain reform, land-use regulation, minimum wage relative to median wage, discrimination statutes, health insurance freedom, telecom deregulation, occupational requirements, miscellaneous regulation, and an aggregate nonfederal regulatory index.

3. Econometric Specification

The primary empirical design is a state fixed-effects panel model. In compact notation, the model can be written as:

$$y_{it} = \alpha_i + \beta X_{it} + \epsilon_{it}$$

In this equation, y_{it} is the economic performance outcome for state i in year t ; α_i is a state fixed effect; X_{it} is the vector of fiscal and regulatory policy variables; and ϵ_{it} is the remaining error term. The model is estimated separately for real GSP per capita and real private-sector GSP per capita.

The fixed effects absorb persistent state-specific factors that do not vary over time. This matters because some states have structural economic advantages or disadvantages that are not directly measured in the dataset. Examples include natural-resource endowments, geography, climate, long-

run migration patterns, and inherited industrial composition. The resulting estimates should be interpreted as within-state associations between policy variables and output, conditional on the included controls and state fixed effects.

Technical note: The cleaned Stata workflow declares the data as a panel with `xtset state_n year` and estimates the two main models using `xtreg ..., fe`. Raw class datasets are not redistributed in the public portfolio package, but the cleaned scripts document the empirical workflow.

4. Selected Results

The results below summarize the strongest and most interpretable findings from the fixed-effects specifications. Coefficients are reported for two outcomes: real GSP per capita and real private-sector GSP per capita. Standard errors are reported in parentheses. Statistical significance is marked as in the original project: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Variable	Real GSP per capita	Real private-sector GSP per capita
State government consumption and investment / income	-230.6* (99.75)	-265.7* (128.3)
State taxation / income	-500.4** (175.2)	-617.0** (225.3)
Sum fiscal policy variables	-54.83** (21.19)	-67.51* (27.25)
Local taxation / income	267.7*** (74.13)	343.0*** (95.36)
Log competing local jurisdictions	8.443*** (2.190)	10.44*** (2.817)
N	1,050	1,050
R-squared	0.452	0.513

Interpretation

The fiscal variables were the most consistent predictors in the models. Higher state government consumption and investment relative to income, higher state tax burden, and the aggregate fiscal policy measure were associated with lower per-capita output in both the total GSP and private-sector GSP specifications. The magnitudes are economically meaningful as well as statistically significant in the reported models.

Some regulatory variables were statistically significant, but many were either statistically insignificant or small in practical terms. For example, minimum-to-median wage ratio and telecom deregulation were not statistically significant in the original specifications. This pattern led to the project's main conclusion: the fiscal-policy variables tested in this model were more consistently associated with state economic performance than most regulatory-policy variables tested.

The positive coefficients on local taxation and competing jurisdictions should be read cautiously. They may reflect omitted features of prosperous metropolitan areas, local public-good provision, or sorting across jurisdictions rather than a simple causal effect of higher local taxation. This is one

reason the project is best interpreted as applied exploratory econometrics rather than a definitive causal estimate.

5. Limitations and Extensions

This project was completed as an undergraduate econometrics paper, so its design has several limitations. First, the model uses state fixed effects but does not fully rule out time-varying omitted variables. Second, policy variables may be endogenous: states with different economic trajectories may adopt different policies rather than policies independently causing those trajectories. Third, some Cato index variables aggregate multiple institutional details, which can make coefficient interpretation less direct. Fourth, the original class workflow did not redistribute raw source datasets in the public portfolio package.

The natural next extensions would be to add year fixed effects, use standard errors clustered by state, test lagged policy variables, evaluate alternative outcome definitions, and explore model specifications that address possible endogeneity. A more advanced version could also separate short-run and long-run relationships or examine whether effects differ by region, industrial structure, or baseline income level.

6. Conclusion

The analysis provides evidence that state fiscal policy variables were more consistently associated with state economic output than the regulatory policy variables included in this project. The findings are consistent with the view that tax burdens, government consumption, and aggregate fiscal posture are important dimensions of state economic performance, while many regulatory variables may require more refined measurement, longer lags, or different identification strategies to evaluate convincingly.

As a portfolio artifact, the project demonstrates an early applied econometrics workflow: building a state-year panel dataset, constructing per-capita outcomes, estimating fixed-effects models in Stata, interpreting regression output, acknowledging limitations, and presenting results to an external economics conference audience.

References

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- Cato Institute. Freedom in the 50 States, seventh edition. William Ruger and Jason Sorens, 2023.
- Regional Economic Analysis Project (REAP). Regional economic data resources.
- U.S. Census Bureau. State population estimates.
- Federal Reserve Bank of St. Louis, FRED. Gross Domestic Product by State release resources.